The Rendering Pipeline

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This Section's Objectives

- What is digital animation composed of?
- How are digital images generated?
- Provide a general overview of the animation pipeline

What is Digital Animation?

- An ordered sequence of digital images

What is a Digital Image?

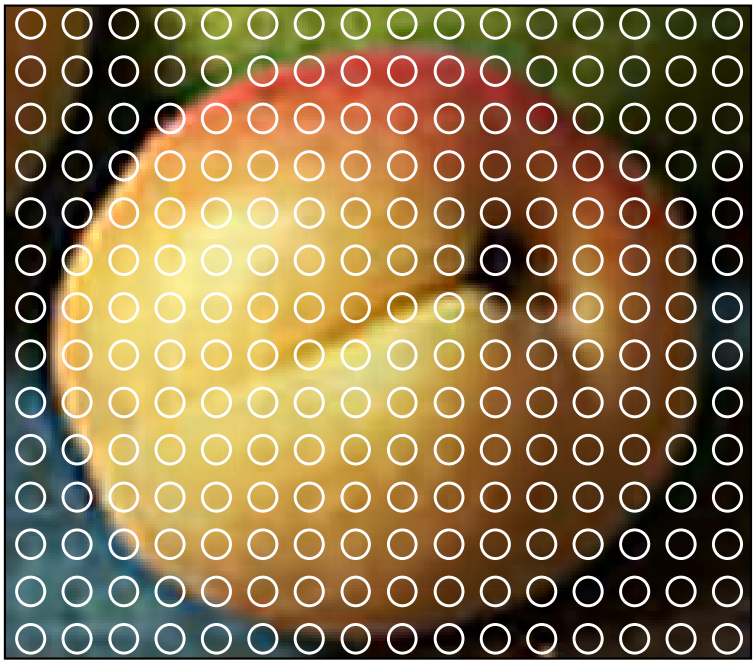
What is a photographic image?

- A two dimensional function
- The value at any point is a colour

What is a digital image?

- just a collection of numbers
- a rectangular array of intensity or color values

What is a Digital Image?



a real image

103 59 12 80 56 12 34 30 1 78 79 21 145 156 52 136 143 65
115 129 41 128 143 50 85 106 11 74 96 14 85 97 23 66 74
23 73 82 29 67 76 21 40 48 7 33 39 9 94 54 19 42 27 6
19 10 3 59 60 28 102 107 41 208 88 63 204 75 54 197 82 63
179 63 46 158 62 46 146 49 40 52 65 21 60 68 11 40 51 17
35 37 0 28 29 0 83 50 15 2 0 1 13 14 8 243 173 161 231
140 69 239 142 89 230 143 90 114 64 29 75 49 24 10 9 5 11
16 9 237 190 82 249 221 122 241 225 129 240 219 126 240 199
93 218 173 69 188 135 33 219 186 79 189 184 93 136 104 65 112
69 37 191 153 80 122 74 28 80 51 19 19 37 47 16 37 32 223
177 83 235 208 105 243 218 125 238 206 103 221 188 83 228 204
98 224 220 123 210 194 109 192 159 62 150 98 40 116 73 28 146
104 46 109 59 24 75 48 18 27 33 33 47 100 118 216 177 98
223 189 91 239 209 111 236 213 117 217 200 108 218 200 100 218
206 104 207 175 76 177 131 54 142 88 41 108 65 22 103 59 22
93 53 18 76 50 17 9 10 2 54 76 74 108 111 102 218 194 108
228 203 102 228 200 100 212 180 79 220 182 85 198 158 62 180
138 54 155 106 37 132 82 33 95 51 14 87 48 15 81 46 14 16
15 0 11 6 0 64 90 91 54 80 93 220 186 97 212 190 105 214
177 86 208 165 71 196 150 64 175 127 42 170 117 49 139 89 30
102 97 41 118 67 24 90 52 16 75 46 16 58 42 19 13 7 9
10 5 0 18 11 3 66 111 116 70 100

a digital image

Pixels = Red, Green, Blue (RGB)

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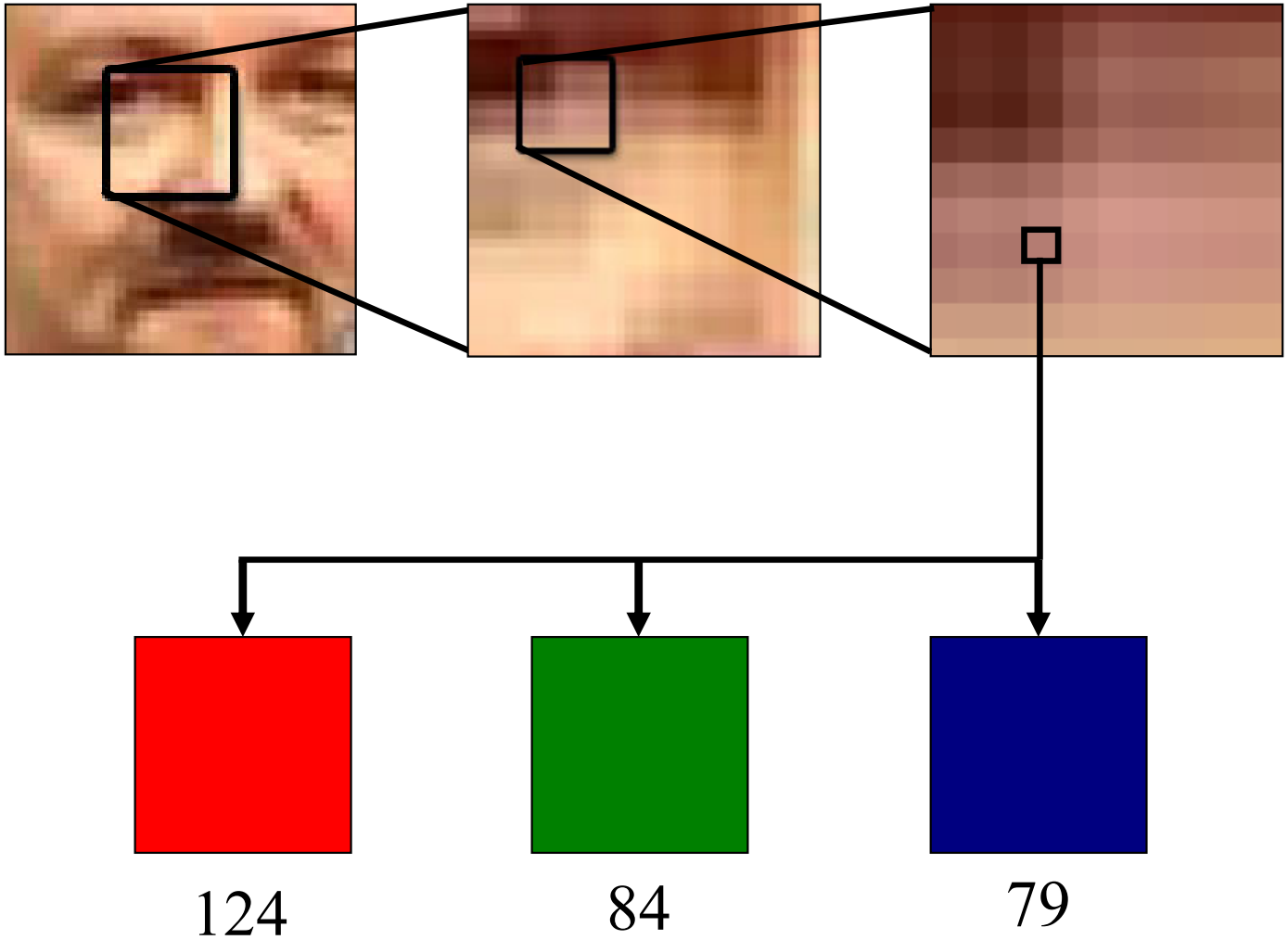
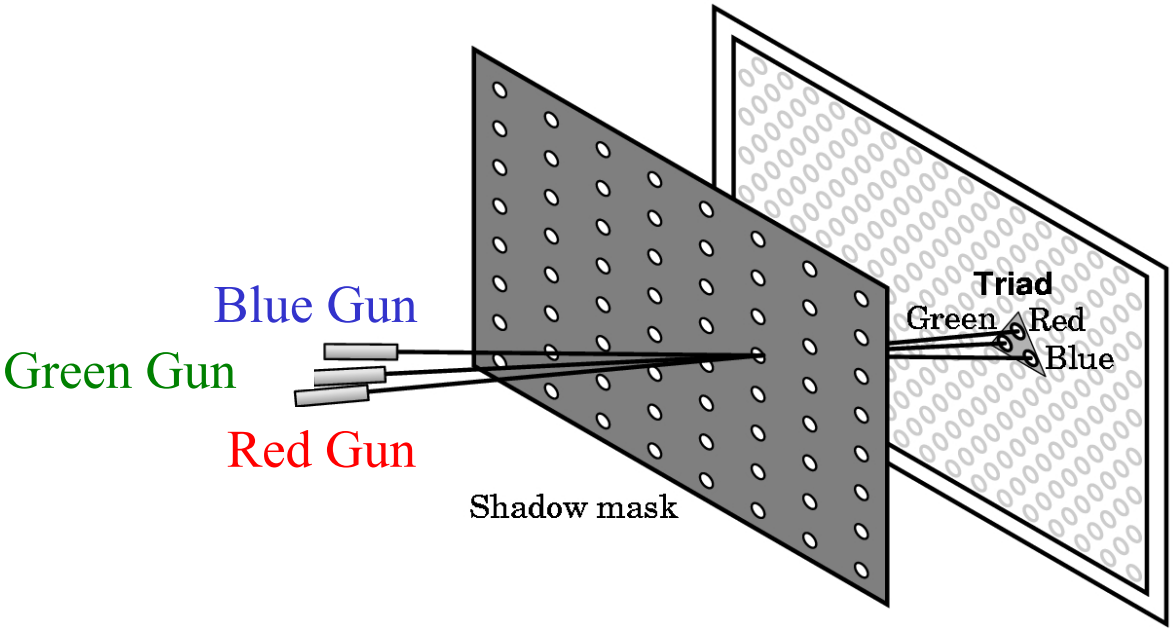
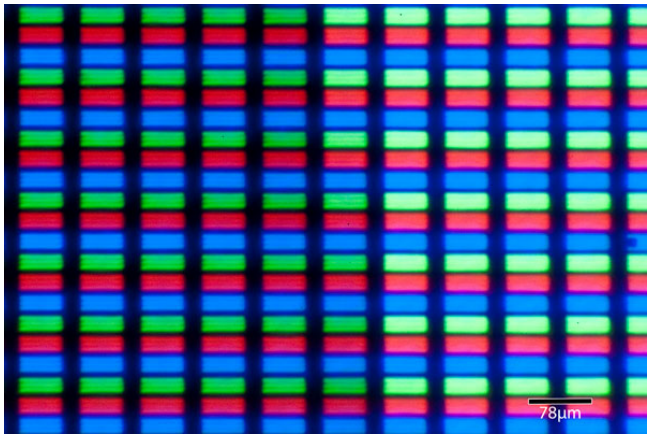


RGB Color on CRTs, LCDs

- In graphics, we normally model light color with 3 values: RGB
- We know that RGB is 'sufficient' since TV and computer screens operate on this principle:





Some Definitions

Frame Buffer – a 2D array of random access memory that stores the color values for each pixel in the displayed image.

Color vs. Greyscale – whether we have 3 separate color channels (RGB) for each pixel or whether we have only one grey level channel.

Resolution – the number of pixels in the frame buffer (w *x* h).

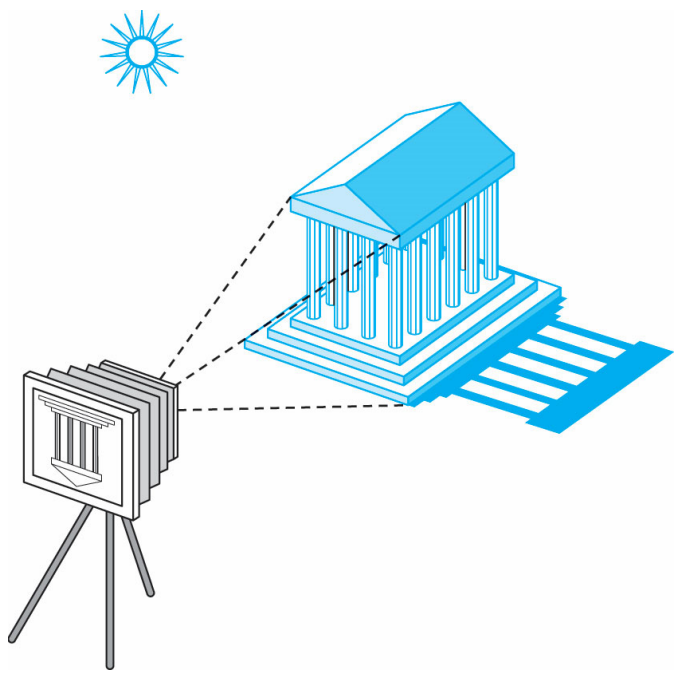
Color Depth – number of bits that are used for each pixel. Determines how many colors can be shown:

- 1-bit depth: only two colors can be shown (i.e. black & white)
- 8-bit depth: allows 2^8=256 colors
- 24-bit depth: 8 bits per color channel, allows millions of colors

The Image Rendering Model

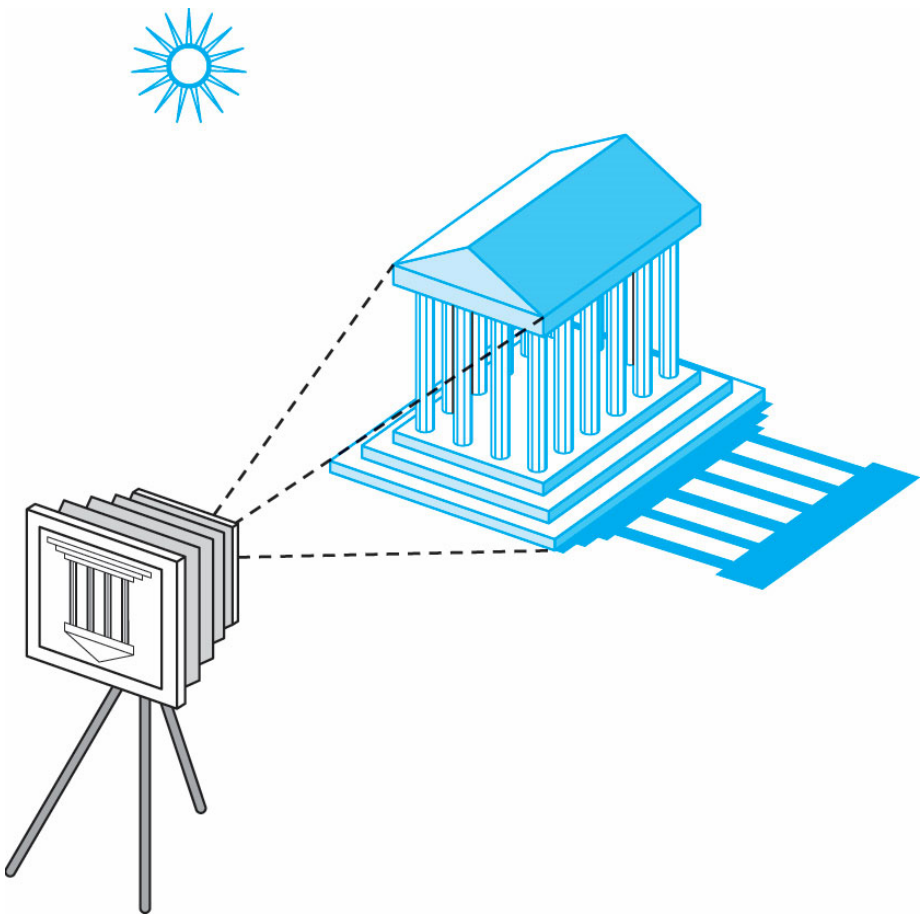
How do we artificially generate these digital images?

- we form two dimensional images using a process analogous to how images are formed by real cameras



The Image Rendering Model

- The Basic Scene Elements
  - Light source(s)
  - Object(s)
  - Viewer
  - Surface materials



Light Source(s)

- Types of Light Sources
  - Point : like the sun, omni-directional
  - Spot : like a flashlight
  - Directional : light at an infinite distance, parallel rays
  - Ambient : the background light level in shadow
  - Area light : like a TV showing white

Object(s)

- Objects are defined as surface geometries
- Types of Objects
  - Point : 2D or 3D vector of float values
  - Line : defined by two points
  - 2D Curve : defined by control points and a method of interpolation
  - Polygon : flat surface defined by its edges
  - 3D Curved Surface : defined by control points and a method of interpolation
  - Volume : array of density values in 3D

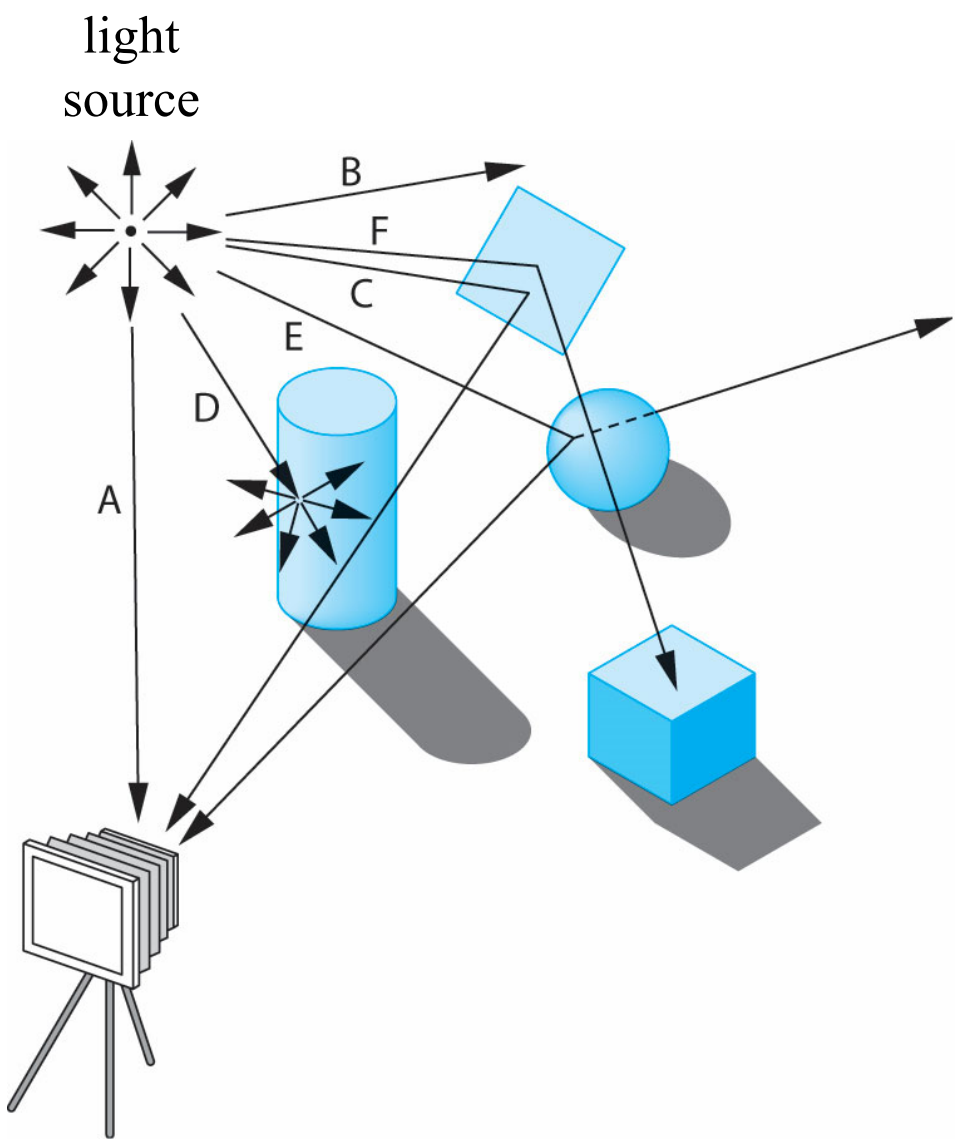
Surface Materials

- Reflective and transmissive properties of surfaces

- Some material attributes
  - Color : basic colour
  - Texture : color varies over surface
  - Specular highlight : shininess
  - Reflectivity : mirror-like
  - Roughness : bumpy surface
  - Transparency : glass-like
  - Translucence : sub-surface scattering

Ray Tracing (not OpenGL)

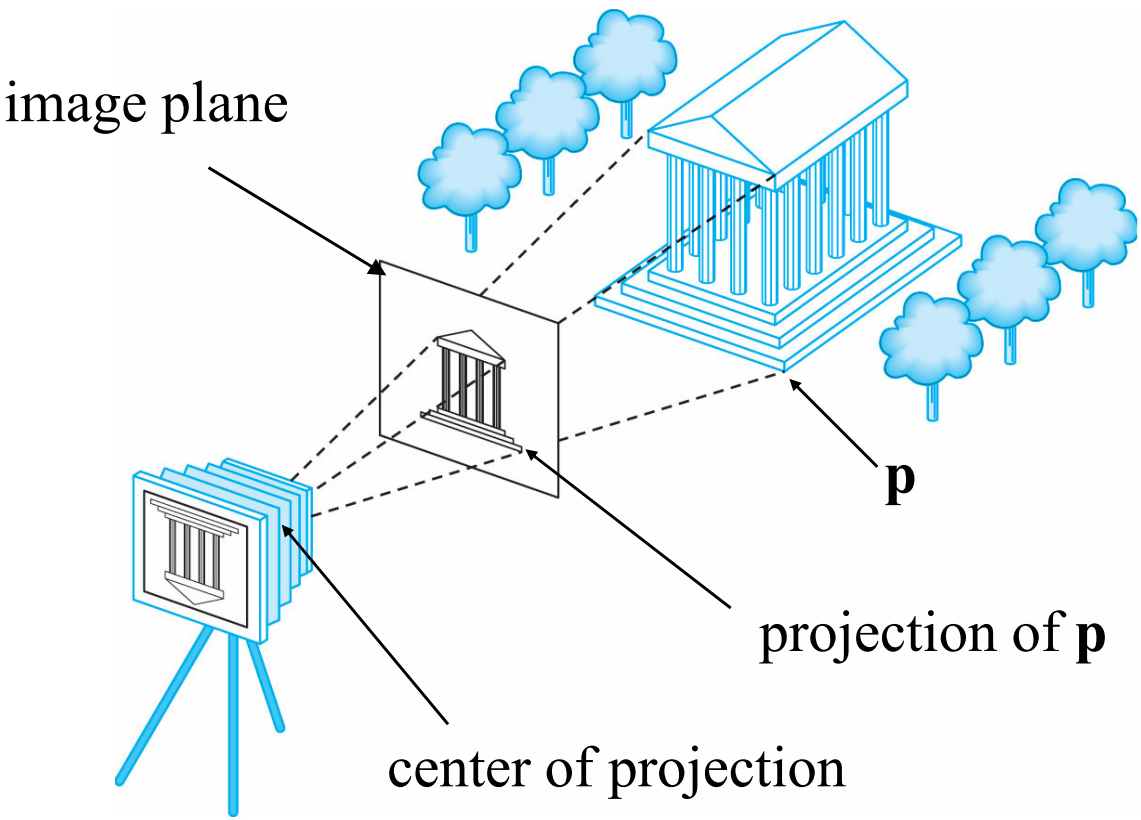
- One simple way to form an image is to follow rays of light (ray trace) from a light source and determine which rays enter the lens of the camera
- Each ray of light may have multiple interactions with objects before being absorbed or going to infinity
- But OpenGL doesn't use ray tracing



Ray Tracing (not OpenGL… yet)

Camera Viewer

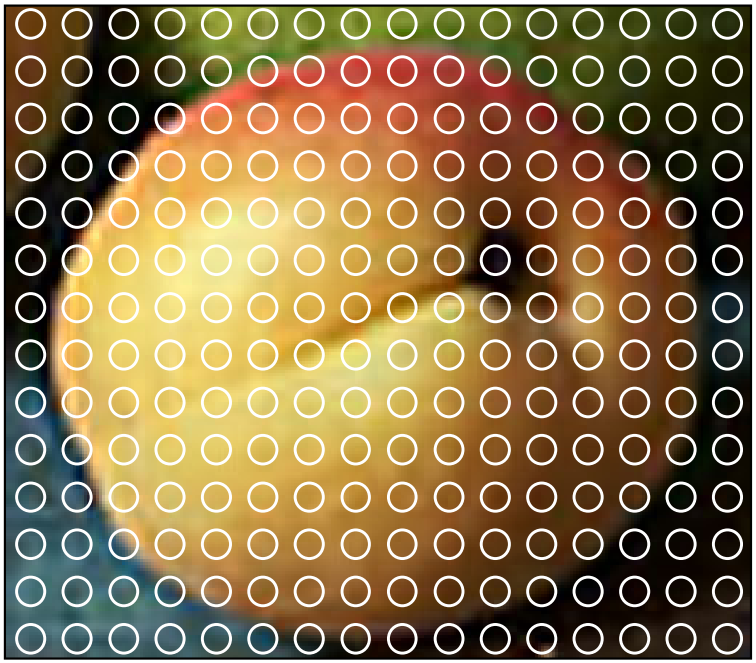
- Synthetic camera



Rasterization

w

h



the image plane

w

103	59	12	80	56	12	34	30	1	78	79	21	145	156	52	136	143	65
115	129	41	128	143	50	85	106	11	74	96	14	85	97	23	66	74	
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19	10	3	59	60	28	102	107	41	208	88	63	204	75	54	197	82	63
179	63	46	158	62	46	146	49	40	52	65	21	60	68	11	40	51	17
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140	69	239	142	89	230	143	90	114	64	29	75	49	24	10	9	5	11
16	9	237	190	82	249	221	122	241	225	129	240	219	126	240	199		
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223	189	91	239	209	111	236	213	117	217	200	108	218	200	100	218		
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93	53	18	76	50	17	9	10	2	54	76	74	108	111	102	218	194	108	
228	203	102	228	200	100	212	180	79	220	182	85	198	158	62	180		
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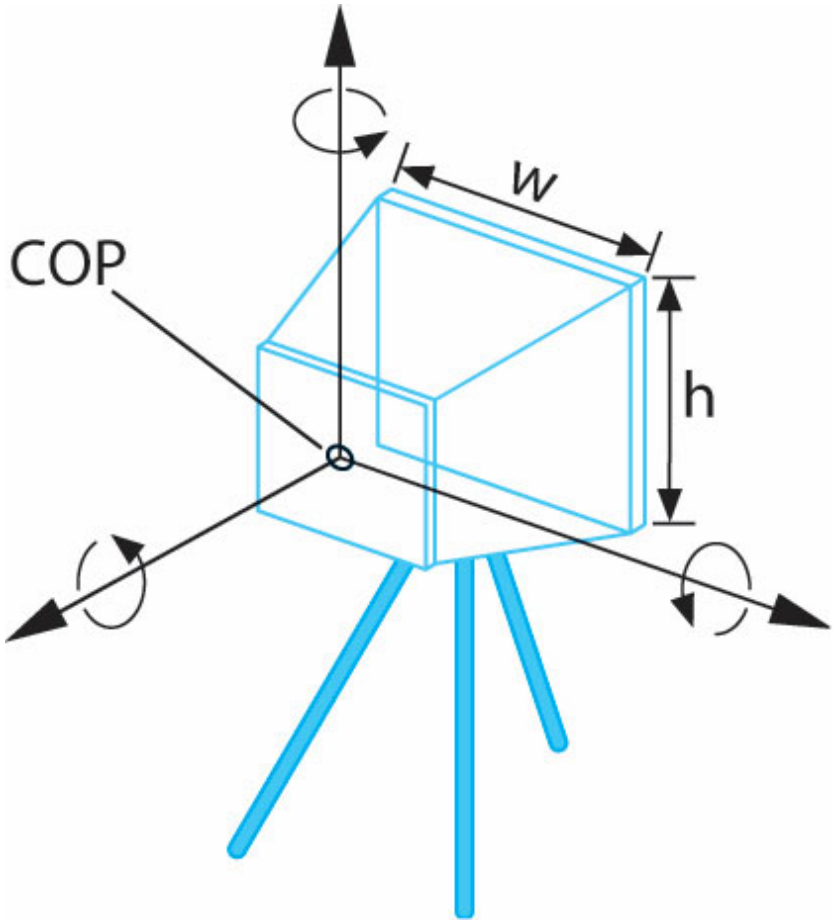
177	86	208	165	71	196	150	64	175	127	42	170	117	49	139	89	30	
102	97	41	118	67	24	90	52	16	75	46	16	58	42	19	13	7	9
10	5	0	18	11	3	66	111	116	70	100							

h

the frame buffer

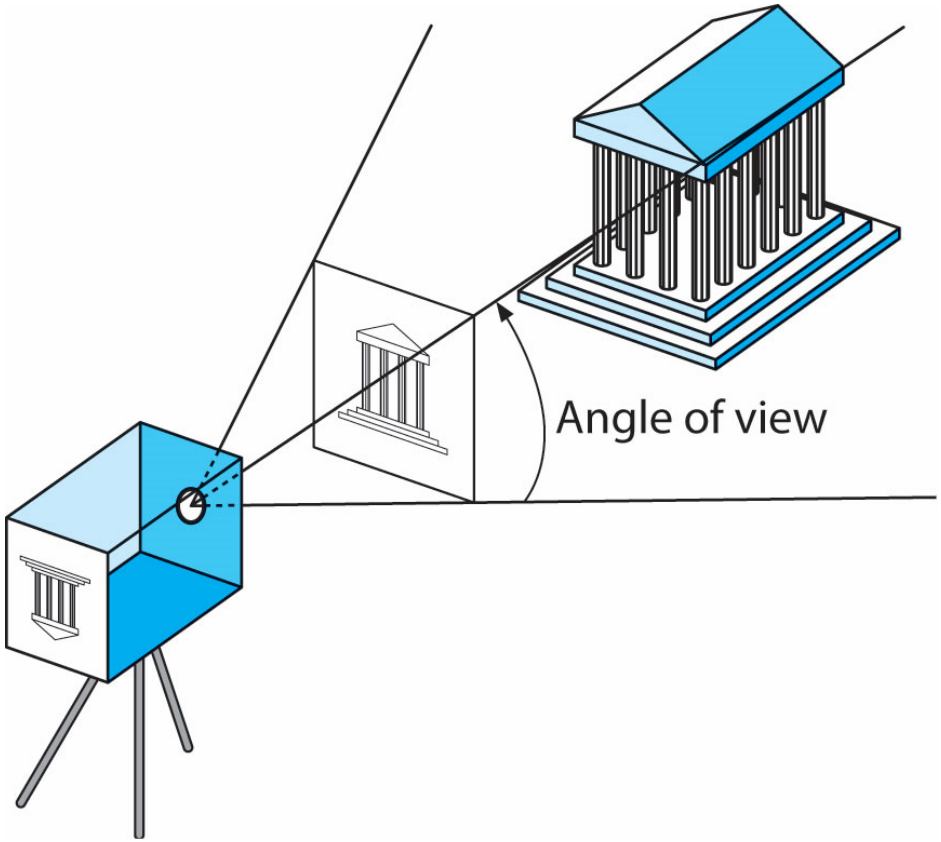
Camera Specification

- Six degrees of freedom
  - Position of center of projection
  - Orientation
- Lens
- Film size (w *x* h)



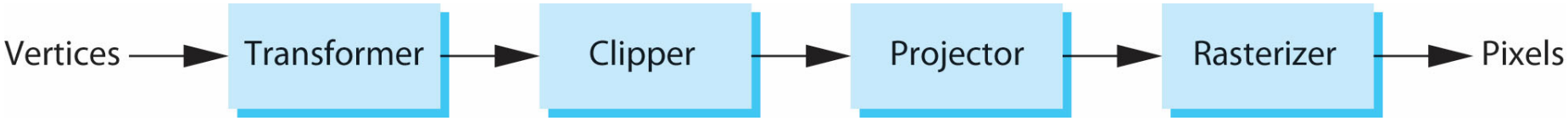
Clipping

- A real camera cannot "see" the whole world
- A virtual camera can't either
- Objects outside field of view are 'clipped'



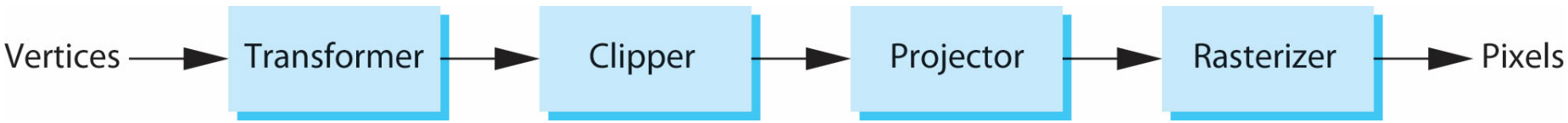
The Image Rendering Pipeline

- All of this put together is called the rendering pipeline:



More Recent Advances – PhysX

All of this put together is called the rendering pipeline:

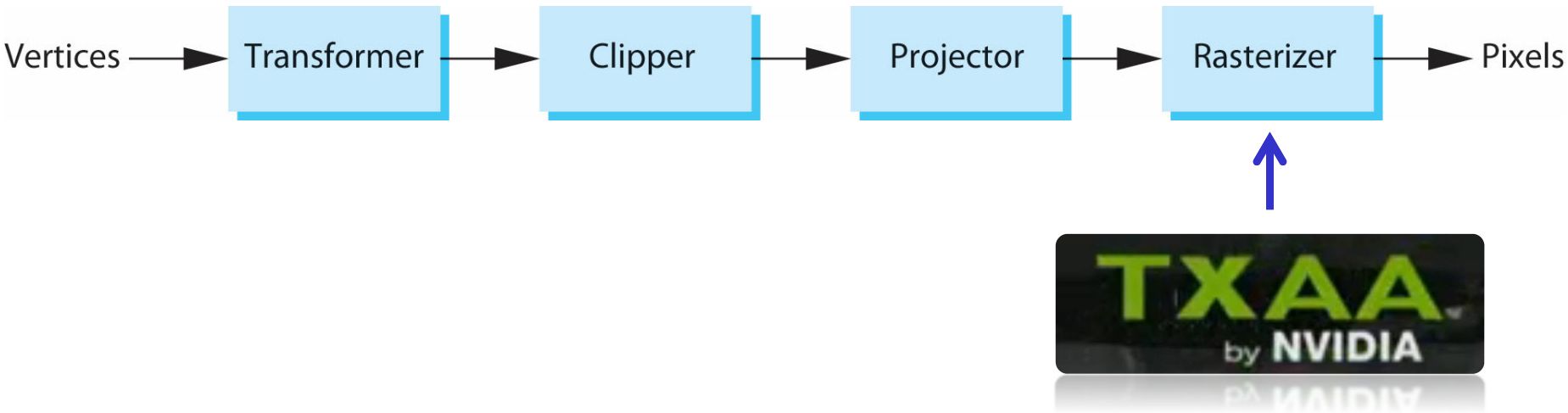


NVIDIA PHYSX

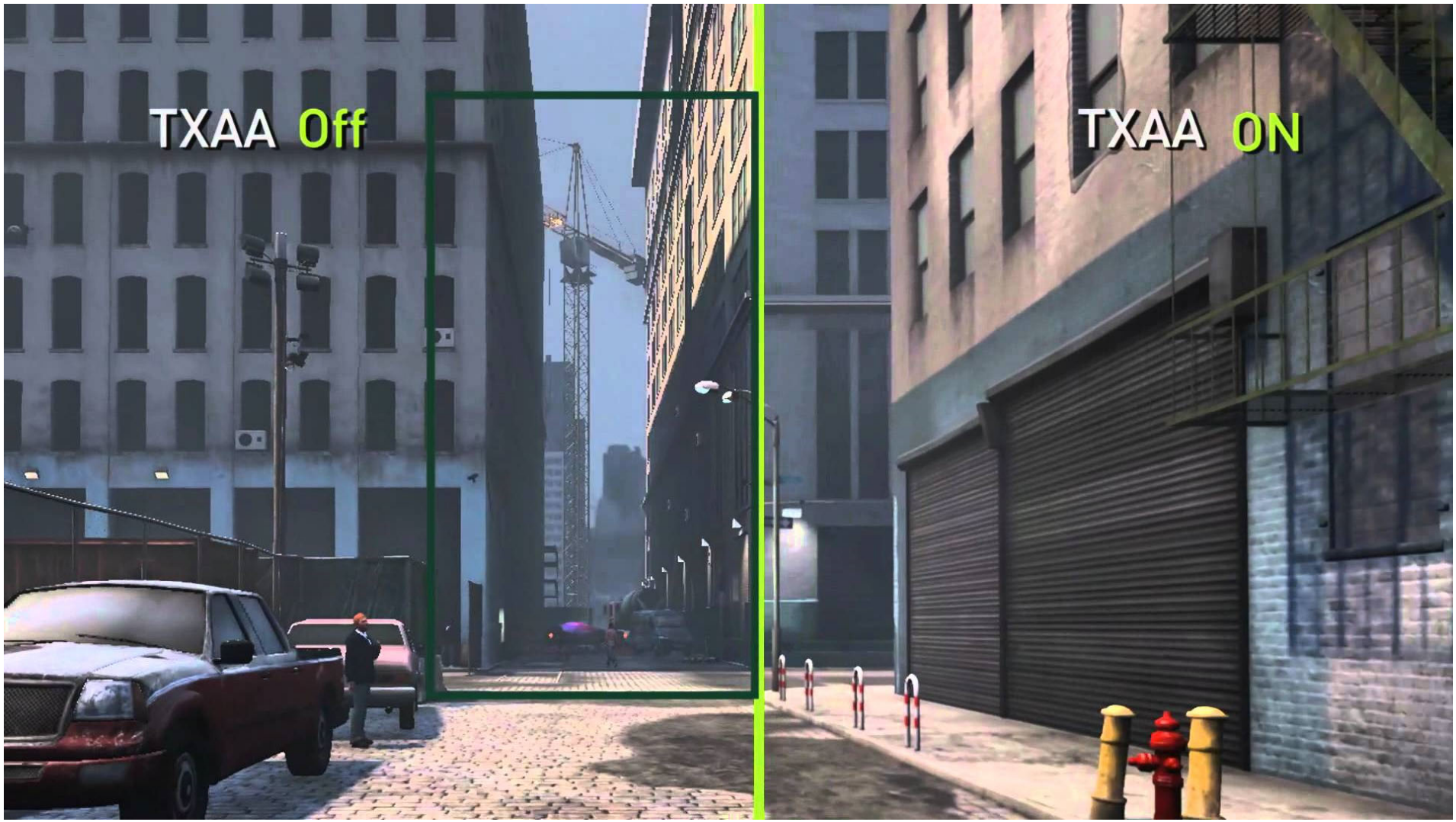
More Recent Advances – PhysX

More Recent Advances – TXAA

All of this put together is called the rendering pipeline:

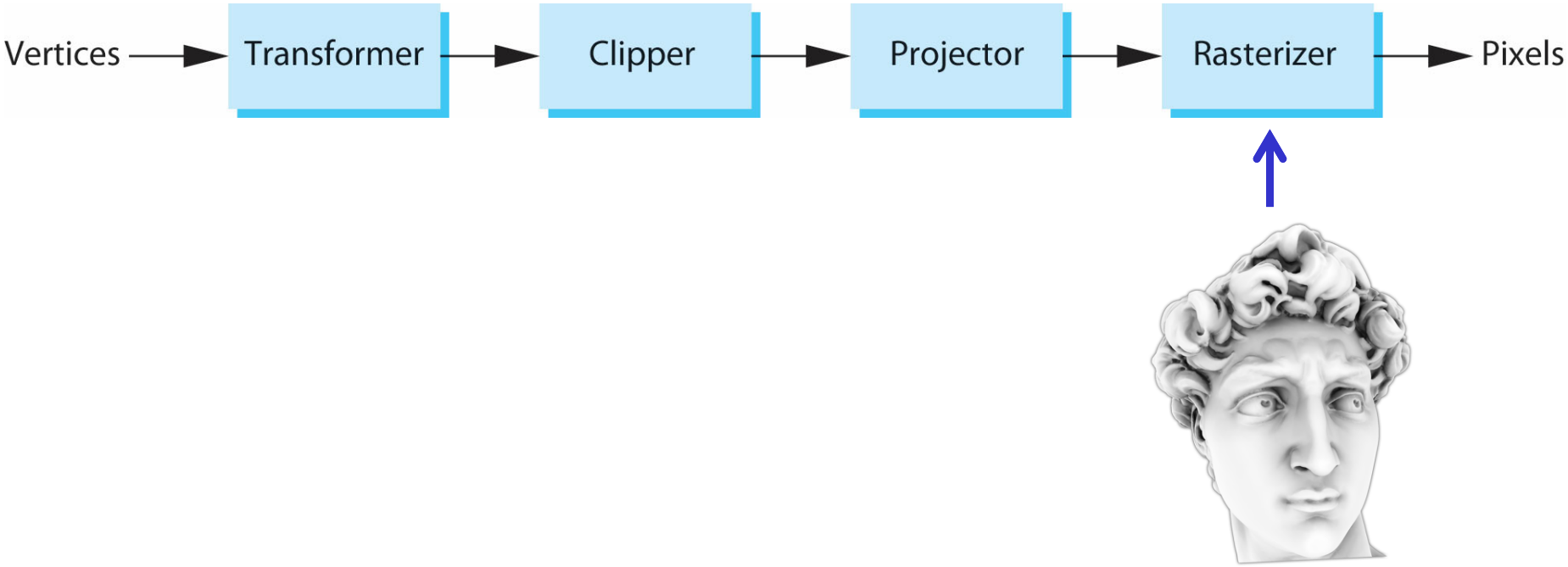


More Recent Advances – TXAA

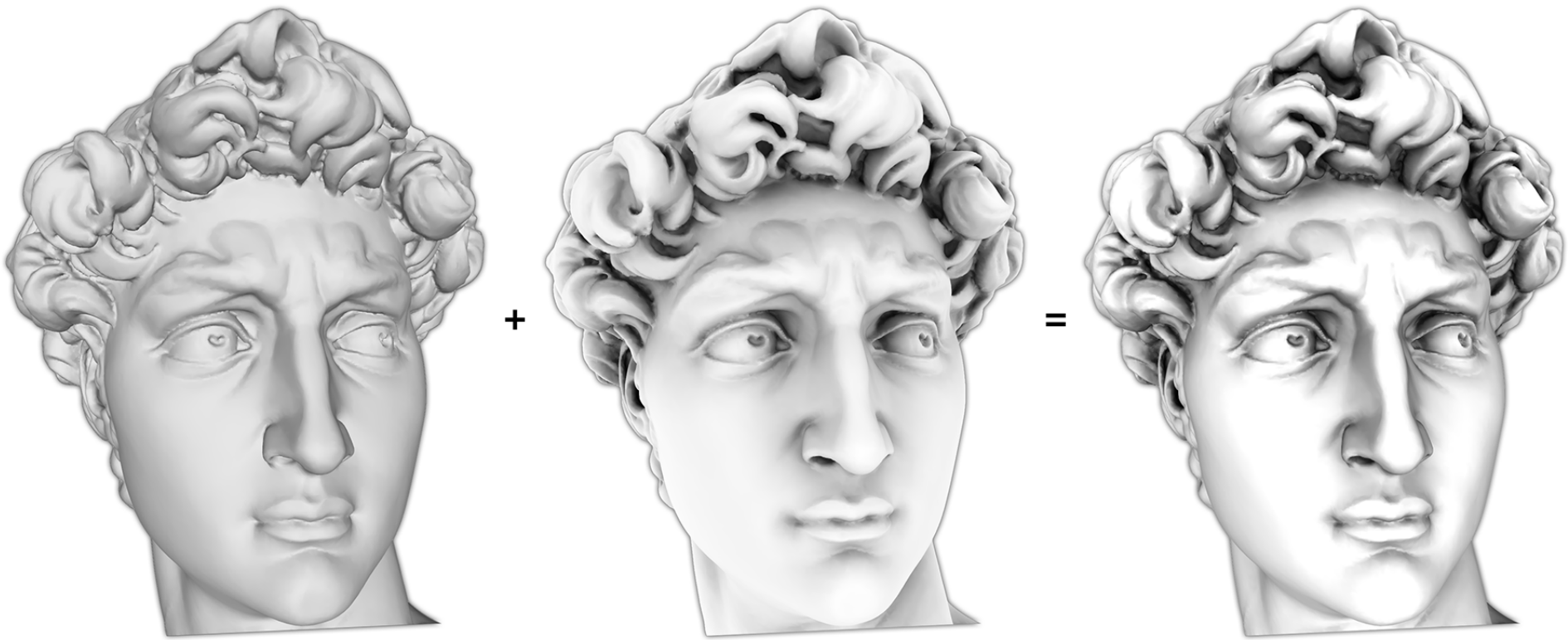


More Recent Advances – Ambient Occlusion

All of this put together is called the rendering pipeline:



More Recent Advances – Ambient Occlusion



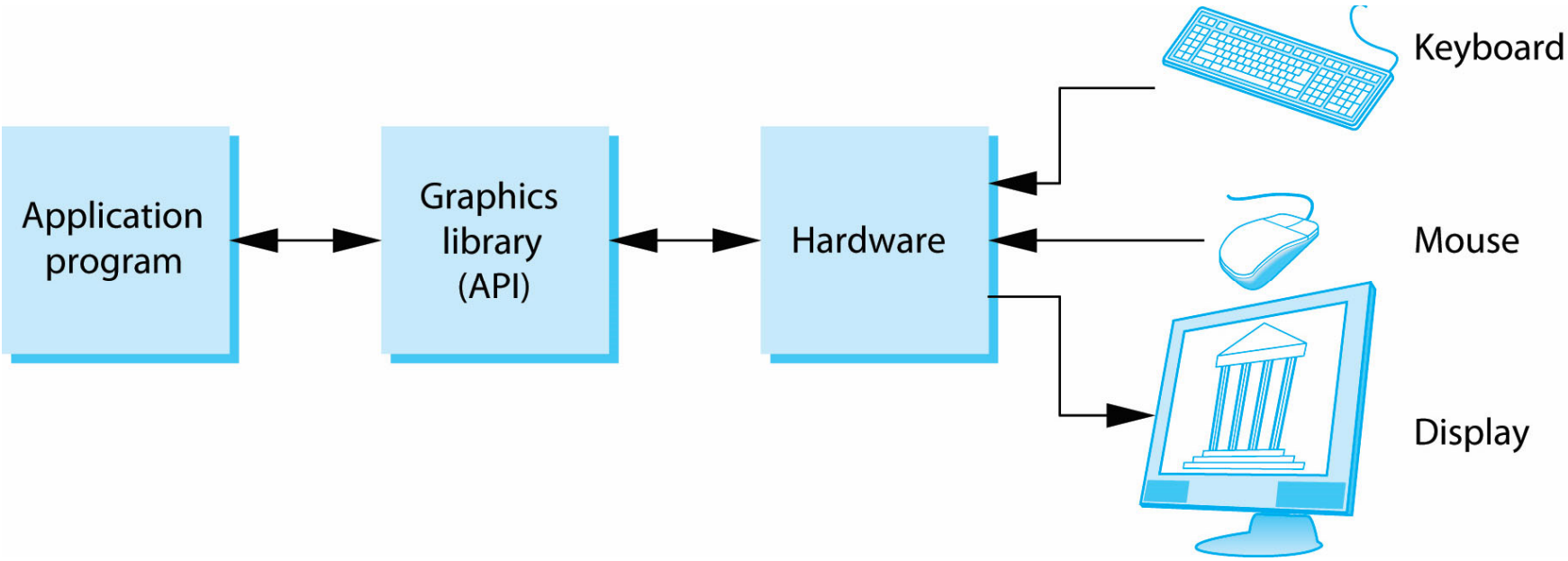
Directional Lighting Ambient Occlusion

The Application Programmer Interface (API)

- We aren't going to:
  - implement this image rendering model ourselves
  - Note: you can learn more about the process of image rendering in the 4th year graphics course
- We are going to:
  - use OpenGL to do the rendering for us
- Our role will be to:
  - specify the objects, camera, material and light properties
  - …and then change them over time – animation

The OpenGL API

- The programmer (you) see:
  - the graphics system through OpenGL
  - user input devices through GLUT



API Contents

- Functions that specify and alter the scene elements
  - Objects
  - Viewer
  - Light Source(s)
  - Materials
- Other information
  - Gathers input from devices such as the mouse and keyboard

Example: Object Specification

- OpenGL supports a limited set of primitives including:
  - Points (1D object)
  - Line segments (2D objects)
  - Polygons (3D objects)
- Also can define some curves and surfaces using the GLU library
  - Quadrics (spheres, cylinders and cones)
- All objects are defined through locations in space called *vertices* (points)

Example Object : Triangle

type of object

```
glBegin(GL_POLYGON)

  glVertex3f(0.0, 0.0, 0.0);
  glVertex3f(0.0, 1.0, 0.0);
  glVertex3f(0.0, 0.0, 1.0);

glEnd( );
```

locations of vertices

end of object definition

Title Slide Images from Canadian Researchers

Image references, clockwise from top left:

- liu, Hsueh-Ti Derek, and Alec Jacobson. “Cubic Stylization.” ACM Transactions on Graphics 38.6 (2019): 1–10.
- Luc Leblanc, Jocelyn Houle, and Pierre Poulin. 2011. Component-based modeling of complete buildings. Graphics Interface. Canadian Human-Computer Communications Society, Waterloo, CAN, 87–94.
- Graham LeBlanc, Andrew Shouldice, Dirk V. Arnold and Stephen Brooks. Multi-band Fourier synthesis of ocean waves. Journal of Graphics Tools, volume 16, issue 2, pp. 57-70, 2012.
- Adrian Blumer, Jan Novák, Ralf Habel, Derek Nowrouzezahrai, Wojciech Jarosz. Reduced aggregate scattering operators for path tracing. Computer Graphics Forum, 35(7):461–473, October 2016.
- Fang Da, Christopher Batty, Chris Wojtan, and Eitan Grinspun. 2015. Double bubbles sans toil and trouble: discrete circulation-preserving vortex sheets for soap films and foams. ACM Trans. Graph. 34, 4, 9 pages.

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